

Business Requirements Document: Hybrid RAG Agent System

Project: Hybrid RAG Orchestration Framework

Version: 1.0.0

Status: Draft

1. Executive Summary

This project aims to implement a highly scalable, hybrid Retrieval-Augmented Generation (RAG) system using specialized AI agents. The system will leverage a semantic vector search combined with keyword-based retrieval to provide precise, context-aware information for enterprise automation.

2. Business Objectives

- Improve retrieval precision by 40% through specialized agent domain mapping.
- Reduce operational latency by utilizing Groq's high-speed LPU inference engine.
- Ensure system modularity for ease of maintenance and independent agent scaling.

3. Functional Requirements

ID	Requirement	Priority
FR-01	Multi-agent orchestration for diverse domain tasks.	High
FR-02	Integration with Groq inference API for real-time response.	High
FR-03	Hybrid indexing support (Vector + BM25).	Medium

4. Technical Specifications

The system will utilize a centralized configuration registry to map domain-specific tasks to optimized open-source models hosted on Groq, ensuring low-latency execution.